Europäisches Patentamt
European Patent Office
Office européen des brevets

SUPPLEMENTARY EUROPEAN SEARCH REPORT

Application Number
EP 22 77 6438

DOCUMENTS CONSIDERED TO BE RELEVANT

Category	Citation of document with indication, where appropriate, of relevant passages	Relevant to claim	CLASSIFICATION OF THE APPLICATION (IPC)
A	MENG-YAO LI ET AL: "Real-time monitoring of antibody glycosylation site occupancy by in situ Raman spectroscopy during bioreactor CHO cell cultures", BIOTECHNOLOGY PROGRESS, AMERICAN CHEMICAL SOCIETY, HOBOKEN, USA, vol. 34, no. 2, 21 January 2018 (2018-01-21), pages 486-493, XP072295789, ISSN: 8756-7938, DOI: 10.1002/BTPR.2604 * abstract * * pages 487-492; figures 1-4 *	1-15	INV. G01N33/68 C12P21/00 A61B5/00 C07K16/06 C12M1/34 C12M1/36 C12N5/00 G05B17/02 G16C20/20 G16C20/70
A	DANIEL A ZAVALA-ORTIZ ET AL: "Interest of locally weighted regression to overcome nonlinear effects during in situ NIR monitoring of CHO cell culture parameters and antibody glycosylation", BIOTECHNOLOGY PROGRESS, AMERICAN CHEMICAL SOCIETY, HOBOKEN, USA, vol. 36, E2924, 14 October 2019 (2019-10-14), pages 1-10, XP072291181, ISSN: 8756-7938, DOI: 10.1002/BTPR.2924 * abstract * * paragraphs [02.1], [03.1], [03.2], [03.3]; figures 1, 2 *	1-15	TECHNICAL FIELDS SEARCHED (IPC) G16C C12M G05B C12P G01N

-/--

The supplementary search report has been based on the last set of claims valid and available at the start of the search.

Place of search	Date of completion of the search	Examiner
Berlin	5 August 2025	Schröder, Gunnar

CATEGORY OF CITED DOCUMENTS

X : particularly relevant if taken alone
Y : particularly relevant if combined with another document of the same category
A : technological background
O : non-written disclosure
P : intermediate document

T : theory or principle underlying the invention
E : earlier patent document, but published on, or after the filing date
D : document cited in the application
L : document cited for other reasons

& : member of the same patent family, corresponding document

1

EPO FORM 1503 03.82 (P04C04)

Europäisches Patentamt
European Patent Office
Office européen des brevets

SUPPLEMENTARY EUROPEAN SEARCH REPORT

Application Number
EP 22 77 6438

DOCUMENTS CONSIDERED TO BE RELEVANT

Category	Citation of document with indication, where appropriate, of relevant passages	Relevant to claim	CLASSIFICATION OF THE APPLICATION (IPC)
A	BRANDON BERRY ET AL: "Cross-scale predictive modeling of CHO cell culture growth and metabolites using Raman spectroscopy and multivariate analysis", BIOTECHNOLOGY PROGRESS, AMERICAN CHEMICAL SOCIETY, HOBOKEN, USA, vol. 31, no. 2, 29 December 2014 (2014-12-29), pages 566-577, XP072293416, ISSN: 8756-7938, DOI: 10.1002/BTPR.2035 * abstract *	1-15	
A	RUTH C ROWLAND-JONES ET AL: "Spectroscopy integration to miniature bioreactors and large scale production bioreactors-Increasing current capabilities and model transfer", BIOTECHNOLOGY PROGRESS, AMERICAN CHEMICAL SOCIETY, HOBOKEN, USA, vol. 37, E3074, 19 September 2020 (2020-09-19), pages 1-14, XP072300068, ISSN: 8756-7938, DOI: 10.1002/BTPR.3074 * abstract * * page 2 * * paragraph [03.7]; figures 8, 9 *	1-15	TECHNICAL FIELDS SEARCHED (IPC)
A	WO 2016/004322 A2 (BIOGEN MA INC [US]) 7 January 2016 (2016-01-07) * abstract; claims; figure 1 *	1-15	

-/--

The supplementary search report has been based on the last set of claims valid and available at the start of the search.

Place of search	Date of completion of the search	Examiner
Berlin	5 August 2025	Schröder, Gunnar

CATEGORY OF CITED DOCUMENTS

X : particularly relevant if taken alone
Y : particularly relevant if combined with another document of the same category
A : technological background
O : non-written disclosure
P : intermediate document

T : theory or principle underlying the invention
E : earlier patent document, but published on, or after the filing date
D : document cited in the application
L : document cited for other reasons

& : member of the same patent family, corresponding document

1

EPO FORM 1503 03.82 (P04C04)

Europäisches Patentamt
European Patent Office
Office européen des brevets

SUPPLEMENTARY EUROPEAN SEARCH REPORT

Application Number
EP 22 77 6438

DOCUMENTS CONSIDERED TO BE RELEVANT

Category	Citation of document with indication, where appropriate, of relevant passages	Relevant to claim	CLASSIFICATION OF THE APPLICATION (IPC)
A	WANG YAN ET AL: "Monitoring Glycosylation Profile and Protein Titer in Cell Culture Samples Using ZipChip CE-MS", JOURNAL OF ANALYTICAL & BIOANALYTICAL TECHNIQUES- 3RD INTERNATIONAL CONFERENCE AND EXHIBITION ON ANALYTICAL & BIOANALYTICAL TECHNIQUESNOVEMBER 22-24, 2012 HYDERABAD INTERNATIONAL CONVENTION CENTRE, INDIA, vol. 08, no. 02, 1 January 2017 (2017-01-01), pages 1-8, XP093302453, ISSN: 2155-9872, DOI: 10.4172/2155-9872.1000359 * abstract * -----	1-15	

TECHNICAL FIELDS SEARCHED (IPC)

The supplementary search report has been based on the last set of claims valid and available at the start of the search.

Place of search	Date of completion of the search	Examiner
Berlin	5 August 2025	Schröder, Gunnar

CATEGORY OF CITED DOCUMENTS

X : particularly relevant if taken alone
Y : particularly relevant if combined with another document of the same category
A : technological background
O : non-written disclosure
P : intermediate document

T : theory or principle underlying the invention
E : earlier patent document, but published on, or after the filing date
D : document cited in the application
L : document cited for other reasons
.....
& : member of the same patent family, corresponding document

1

EPO FORM 1503 03.82 (P04C04)

ANNEX TO THE EUROPEAN SEARCH REPORT ON EUROPEAN PATENT APPLICATION NO. EP 22 77 6438

This annex lists the patent family members relating to the patent documents cited in the above-mentioned European search report. The members are as contained in the European Patent Office EDP file on
The European Patent Office is in no way liable for these particulars which are merely given for the purpose of information.

05-08-2025

Patent document cited in search report	Publication date	Patent family member(s)	Publication date
WO 2016004322 A2	07-01-2016	EP 3164481 A2	10-05-2017
		US 2017130186 A1	11-05-2017
		US 2020392447 A1	17-12-2020
		WO 2016004322 A2	07-01-2016